Abstract

This project developed a machine learning model to predict the critical temperature (Tc) of superconductors using datasets containing elemental properties and material compositions. We evaluated multiple regression models and performed feature importance techniques, including SHAP to identify the most impactful predictors . We also employed anomaly detection methods (SVM and Isolation Forest) that highlighted potential outliers. Finally, To explore novel superconductors, we combined compositional data with Bayesian Optimization, applying scientific constraints to guide the search. This approach demonstrates the potential of machine learning to accelerate discoveries in superconductivity.

Introduction

Superconductors

Superconductors are materials that exhibit no electric resistance when they are cooled down to a specific critical temperature. During the recent years several advancements have been made in the field of superconductivity in general. Constantly scientists come up with novel combinations of already existing materials in order to achieve higher critical temperatures. Finding superconductors with a high enough critical temperature, even room temperature, would be a scientific and technological breakthrough. This is because it will enhance several aspects of modern technology such as electronics, power transmission, energy storage, and even healthcare. In an attempt to assist such research efforts, data analysts throughout the world in collaboration with material scientists are trying to come up with a model that predicts combinations of elements in order to achieve a high enough critical temperature, making the superconductors more practical to use in everyday aspects. The most important factors that determine the superconductivity of a material revolve around the electron structure of it, its crystal structure and its natural electronic properties. In this project we are going to develop a model that does the above from solely a data analysis point of view. Meaning that further restrictions should be applied based on material science and chemistry knowledge so far.

Dataset

The dataset we found represents a comprehensive collection of material compositions and their properties . It consists of two csv files (train.csv , unique_m.csv) , both files includes 21,263 entries, where train file consists of 82 features capturing statistical descriptors of elemental properties such as atomic mass, first ionization energy, atomic radius, density, and thermal conductivity and the target variable, critical_temp, specifies the superconducting transition temperature in Kelvin. While unique_m provides detailed compositional data, with each element's fractional presence in a material explicitly recorded as a numeric value, where the column names correspond to the chemical elements (e.g., H, O, Cu).

Methodology

Model training

The first part of our project focuses on training a model to predict the critical temperature (Tc) of materials using the features provided in the first file of the dataset (train.csv).

We began by splitting the dataset into training and testing subsets using scikit-learn, with 80% allocated for training and 20% for testing. We standardized the data for all the models we will test for efficiency. While not all models require scaling, it improves performance for some (e.g., SVR, Ridge) and does not harm scale-invariant ones.

We continued by training and evaluating a range of machine learning models and comparing their performance using root mean squared error (RMSE) as the evaluation metric (we also included R^2 score but our decisions were based on RMSE). We started from simple regression models like Linear Regression, Random Forest, and Gradient Boosting, and continued with more advanced techniques: Ridge Regression, Lasso Regression, Support Vector Regressor, K-Nearest Neighbors, Decision Tree, and a Multilayer Perceptron. Initially, all models were trained with their default parameters in order to establish baseline performance and cross-validation was performed to obtain more robust performance estimates.

The best performing model based on default parameters is Random Forest Regressor which we then proceeded to tune its parameters using GridSearchCV. To tune the model we adjusted the number of estimators, maximum tree depth, and minimum samples required to split a node. The grid search was performed using five-fold cross-validation to identify the optimal combination of parameters.

With this method we obtained the best parameters for the lowest RMSE score.

Feature extraction

The file we used to train the above model included 82 features. This can lead to it being computationally expensive while some features might not impact the model positively. In order to fix this we performed a feature importance

First used the built-in feature importance provided by the Random Forest model. Using the optimal parameters we found we extract the top impactful features. We continued by trying to find a good balance between accuracy and minimize computation cost. .To achieve this, we tested our model using different combinations of the top-performing features (top: 5,10,15,20,30,50,70,82) , evaluating how each set impacted our models accuracy.

Due to the complexity of our dataset we also decided to use a more advanced feature importance technique called SHAP (SHapley Additive exPlanations). SHAP helps explain how each feature contributes to the model's predictions by assigning a value to each feature's impact on a specific prediction. These values show whether a feature increases or decreases the predicted outcome and by how much. Using shap we extracted a ranked feature importance but also but also gained detailed insights into how each feature impacts individual predictions. Finally we tested our model for different combinations of the top-performing features but this time based on SHAP values.

Anomaly Detection

After the completion of feature engineering and the first visualizations of the data we noticed many datapoints that do not seem to follow the main trend (around 1% to 2%). This could be due to several reasons. Maybe our database contains false input about some elements or combinations of them. Or it could be that their mere nature exhibits extraordinary properties and causes them to behave in a different way than the majority.

In either case in order to confirm such speculations the intervention of an expert is required to add scientific restrictions that could pinpoint any hypothetical issues. This is the reason we decided to perform anomaly detection techniques.

The first approach using a Support Vector Machine that will track anomalies in respect of all the features contained in the dataset.

In the second approach we are going to use an Isolation Forest in order to track anomalous activity regarding critical temperatures (critical_temp) in respect to the feature of Mean Atomic Mass (mean_atomic_mass). The reasoning behind this is to assist users of the model to determine whether the critical temperature of the datapoints is in agreement to that of the current research available and thus eliminate the possibility of false input in the dataset.

- **One Class Support Vector Machine Approach**

The way that SVMs work is by representing each datapoint as a vector whose components match the values of the features associated with it. In our case each vector will have 82 components. Next it searches for a hyperplane that will contain the majority of the normal datapoints, namely the ones that follow the underlying trend. Once this hyperplane is found, any datapoints that don't belong in it are classified as anomalies and are treated as such. In our case we are interested in the classification between anomalous and non-anomalous datapoints this approach yielded seemingly acceptable results in respect to the restriction that we set (1%).

However, the high dimensionality of our problem posed a problem regarding the interpretation of these results via the means of visualization. In order to tackle this difficulty we used some dimensionality reduction techniques such as Principal Component Analysis and t-Distributed Stochastic Neighbor Embedding. These techniques proved fruitful and provided us with graphs that make it easy to distinguish the anomalies from the normal points.

This approach however makes it difficult for user future users of the model to make assumptions from the results due to the complex relation of the features assigned to each specific element or combination of elements. This is the reason that we deemed it necessary to use the second approach. An approach that will help users make assumptions about the anomalous nature of the data based on a single feature. A feature that will make cross checking between accepted scientific values of the data and the data itself easier. By doing this they will be able to achieve anomaly detection that is based on further scientific restrictions.

- **Isolation Forest**

Isolation Forest's primary operation is to construct a multitude of Isolation Trees that partition the data randomly among the feature space of Mean Atomic Mass (**mean_atomic_mass**). Due to the nature of anomalies, being isolated and differently distributed than the majority, the algorithm isolates them easily with far less splits than the ones required to isolate the normal datapoints. Then it computes the average isolation paths across multiple trees in order to identify the outliers based on their ease of isolation mentioned above.

Our goal here is to isolate the datapoints that do not seem to behave the same as the majority does in order to make users able to do the cross checking mentioned before.

Material synthesis

The last part of our project contains a model that is able to predict new superconductors with relatively high critical temperature. In order to achieve this we used the second file (unique_m.csv) of the dataset which contains the material composition of each datapoint from the first file. Merging those two datasets we get a new one containing both the features from the first and its composition represented as numeric values, with each column corresponding to a chemical element.

Then we trained a RandomForestRegressor model based on the new dataset. In this section we adjusted the parameters of the model slightly from the previous configuration to reduce computational cost while maintaining good performance.

Because iterating through all the possible combinations would be computationally impossible we needed to employ an algorithm that could efficiently explore and optimize the search space while minimizing the number of evaluations required. For this purpose we decided to use a Bayesian optimization algorithm

Bayesian Optimization (BO) is a probabilistic method for optimizing black-box functions that are expensive to evaluate. It works by building a surrogate model to approximate the objective function and iteratively selecting the most promising points to evaluate based on an acquisition function, which balances exploration of the search space and exploitation of known high-performing regions.

We defined a search space with binary variables representing the presence (1) or absence (0) of each chemical element in the composition.

Using Bayesian Optimization, we iteratively refined the search by predicting the critical temperature (Tc) of each proposed composition with the trained RandomForest model The optimization process focused in finding the most promising candidates based on predicted Tc, guided by an acquisition function balancing exploration of new compositions against refinement of high-performing ones. This was achieved by defining an objective function that evaluated each proposed composition. For this method to produce realistic results we would need numerous theoretical and practical constrains to guide the optimization process because obviously most random compositions produced will be either scientifically invalid or impractical for synthesis. For this project we just employed two : limiting the maximum number of elements in a composition and enforcing the presence of at least one transition metal which is found in almost all superconductors. These constraints were implemented through a penalty system in the objective function. Compositions exceeding the maximum number of allowed elements were penalized proportionally to the excess, while those lacking a transition metal incurred a significant fixed penalty. This system ensured the optimization process focused on more scientifically plausible and experimentally feasible material compositions.

The objective function returned the negative predicted Tc because Bayesian Optimization minimizes the objective function, hence focusing on maximizing Tc while adhering to the imposed constraints.

We tried the running the algorithm couple of times changing the ncalls ,which represents the number of function evaluations that the optimization algorithm will perform, and the max components limitation in order to find a good result.

Results

This section presents the outcomes of our machine learning analysis done with the methods described above.

Model training

As we can see from Table 1 and Table 2 Random provided the best performance in terms of RMSE=9.04 and cross-validation RMSE=9.79 . The Gradient Boosting, Decision Trees, K-Nearest Neighbors and MLP algorithms revealed mediocre performance. Linear Regression, Ridge, and Lasso were struggling with overall high RMSEs. SVR showed some potential but overfitted , considering higher cross-validation RMSEs. Overall, Random Forest is the most reliable model.

Model	Linear Regression	Random Forest	Gradient boost	K-Nearest Neighbors
RMSE	17.38	9.04	12.33	10.44
Cross validation RMSE	17.70	9.79	12.76	11.35

Table 1range of model performances evaluated by RMSE metric

Model	Ridge Regression	Lasso Regression	SVR	Decision Tree	MLP
RMSE	17.40	18.13	15.78	11.63	10.63
Cross validation RMSE	17.71	18.52	16.60	12.67	11.49

Table 2 performances of more advanced models evaluated by RMSE metric

With hyperparameter tuning, Random Forest's performance further improved, achieving an RMSE of 8.95 using optimized parameters: {max_depth=20, min_samples_split=2, n_estimators=300}

Feature extraction

From Figure 1, we see that **range_ThermalConductivity** and **wtd_gmean_ThermalConductivity** are the most influential features deducted from the Random Forest feature importance. This was expected since Thermal conductivity is highly correlated with the critical temperature. Figure 2 further supports this, showing **range_ThermalConductivity** as the feature with the strongest positive SHAP impact on Tc, while other features like **mean_Density** and **std_atomic_mass** exhibit more varied effects.

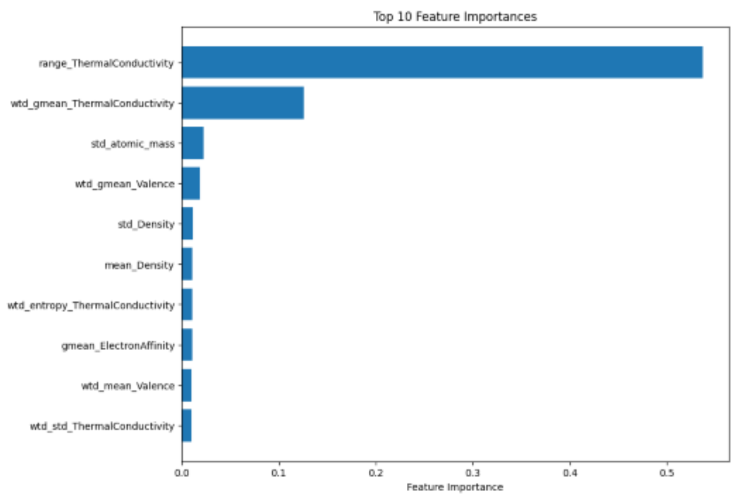

Figure 1. Top 10 Feature Importances Chart

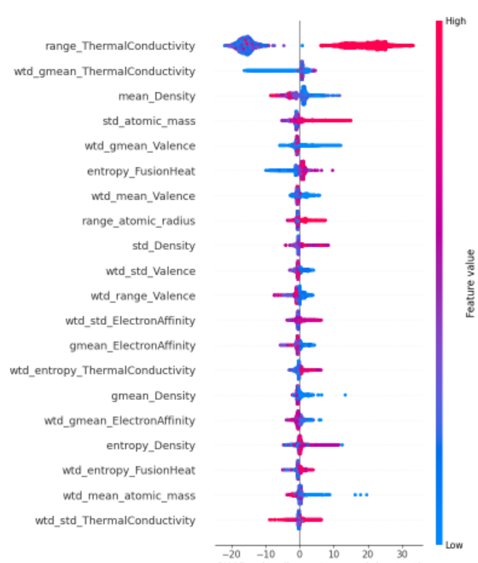

Figure 2. SHAP Summary Plot

Testing the model for different top feature inclusions, we gained additional insights into the importance of our features.

Figure 3 and Figure 4 compare RMSE as more features are added. Random Forest (Figure 3) performs exceptionally well with the top 20 features, achieving a lower RMSE than SHAP (Figure 4), showing its strength in identifying the most important predictors early. However, beyond the top 20, Random Forest makes only small improvements while SHAP shows a smoother, more consistent decrease in RMSE, which stabilizes around 50 features. This points to the fact that Random Forest is excellent in feature selection, giving the top features fast, while SHAP is more reliable in larger feature sets.

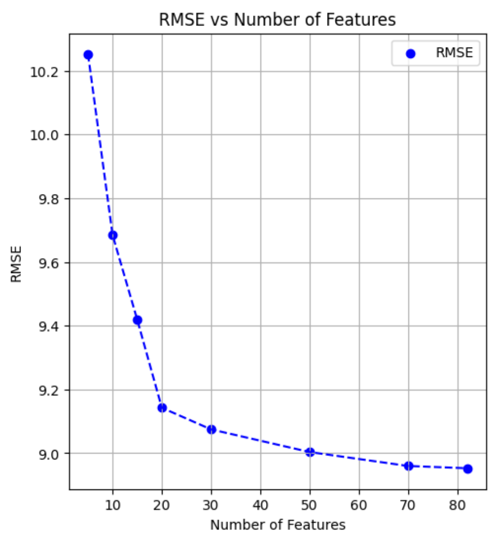

Figure 3. RMSE for different inclusions of top features identified by Random Forest

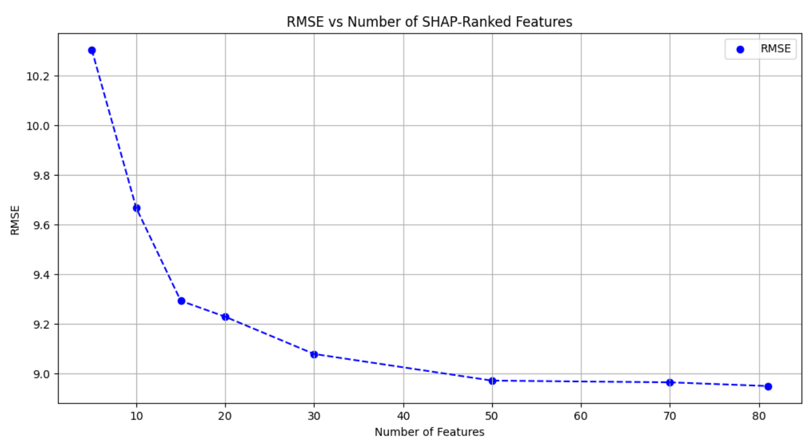

Figure 4. RMSE for different inclusions of top features identified by shap

Overall, an extra cost-efficient approach would use Random Forest's top 20 important features, while SHAP's top 50 features provide a more balanced and reliable model.

In conclusion for this part of the project the model we found to be accurate and cost efficient is Random Forest Regressor with optimized parameters (max_depth=20, min_samples_split=2, and n_estimators=300) and using only the top 50 important features calculated by shap.

Anomaly Detection

SVM: With the implementation of the SVM algorithm we managed to track 215 anomalies, roughly 1% of the original dataset. After the application of dimensionality reduction with the use of PCA and t-SNE we created the following graphs.

The presence of overlapping datapoints shown in Figure 5,6 could suggest that some anomalies are not well distinguished from the rest of the normal data by the SVM model

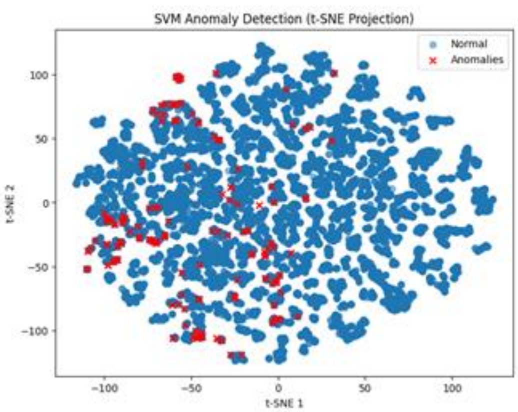

Figure 5. SVM anomaly detection using t-SNE

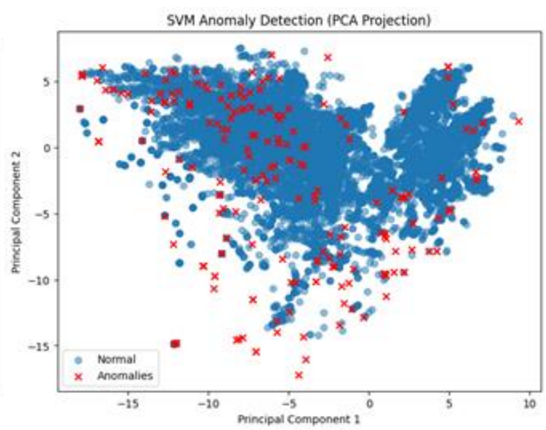

Figure 6. SVM anomaly detection using PCA

Isolation Forest: The use of the Isolation Forest algorithm managed to yield 213 anomalies. Below lie the plots of the original dataset and the one that highlights the tracked anomalies.

From the Figure 7 it appears that the anomaly detection was executed in the proper manner regarding the limitation of roughly ~1% anomalous points. However, some limitations based on the nature of the algorithm are visible. Such as the fact that it is unable to designate anomalies near clustered data. Also the fact that the anomalies tend to be clustered for a Mean Atomic Mass around 200 may indicate that there may be sub-groups of anomalies that feature common characteristics although further scientific analysis must be done in order to validate this claim.

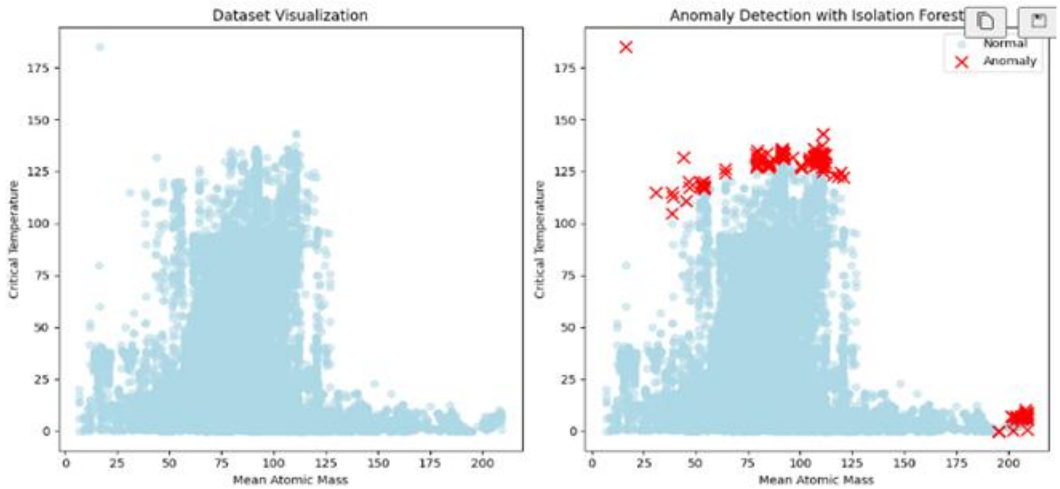

Figure 7. anomaly detection using isolation forest

Material synthesis

The surrogate model trained on the combined data file , as we would expect , perform worse with an RMSE 10.4 since the trends that underline chemical composition and critical temperature are way more complicated.

First we ran the optimization for ncalls = 300 and max components =4 and retrieved the results:

Chemical Formula of Optimal Material:	**HAlArMo**
Predicted Tc:	**33.24662848719903**

With a convergence plot shown in Figure 8

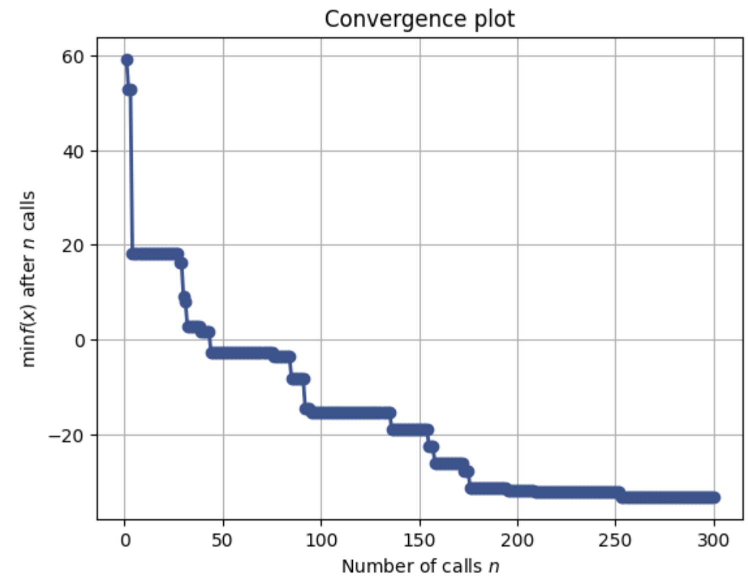

Figure 8. Convergence plot for ncalls = 300 showing to stabilize at n = 180

The n call for this case showed to stabilize at around 180 so we tried again this time with ncall = 180 and max components =3 retrieved:

Chemical Formula of Optimal Material:	**HeMnZn**
Predicted Tc:	**32.10312848719903**

Unfortunately these two formulas have not been researched at all and our predictions cannot be verified (for now). Of course, the scientific validity of this model is limited, as the training process does not account for the quantities of each element present, nor does it incorporate significant scientific guidance or constraints.